Alma Mater Studiorum · Universitá di Bologna

Department of Computer Science and Engineering
Master Degree in Artificial Intelligence

Machine Learning Project

Bank Marketing Binary Classification

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Project Objective

- Predict whether a client subscribes to a term deposit.
- Binary classification on a bank marketing dataset.
- Goal: maximize ROC-AUC score on Kaggle leaderboard.
- Dataset size:
  - Train: 750k rows, 18 features
  - Test: 250k rows

Dataset Description

Features include:

- Client data: age, job, marital status, education
- Financial info: balance, housing loan, personal loan
- Campaign info: duration, pdays, campaign, previous contacts

Target:

- y = subscription to term deposit

Dataset Preview

First 5 rows of the Bank Dataset

	id	age	job	marital	educ.	def.	bal.	hous.	loan	cont.	day	mon.	dur.	cam.	pday	prev.	pout.	y
0	0	42	techn.	married	sec.	no	7	no	no	cell.	25	aug	117	3	-1	0	unk.	0
1	1	38	blue-c.	married	sec.	no	514	no	no	unk.	18	jun	185	1	-1	0	unk.	0
2	2	36	blue-c.	married	sec.	no	602	yes	no	unk.	14	may	111	2	-1	0	unk.	0
3	3	27	student	single	sec.	no	34	yes	no	unk.	28	may	10	2	-1	0	unk.	0
4	4	26	techn.	married	sec.	no	889	yes	no	cell.	3	feb	902	1	-1	0	unk.	1
5	5	24	admin.	single	sec.	no	1882	yes	no	cell.	20	apr	1010	3	-1	0	unk.	0

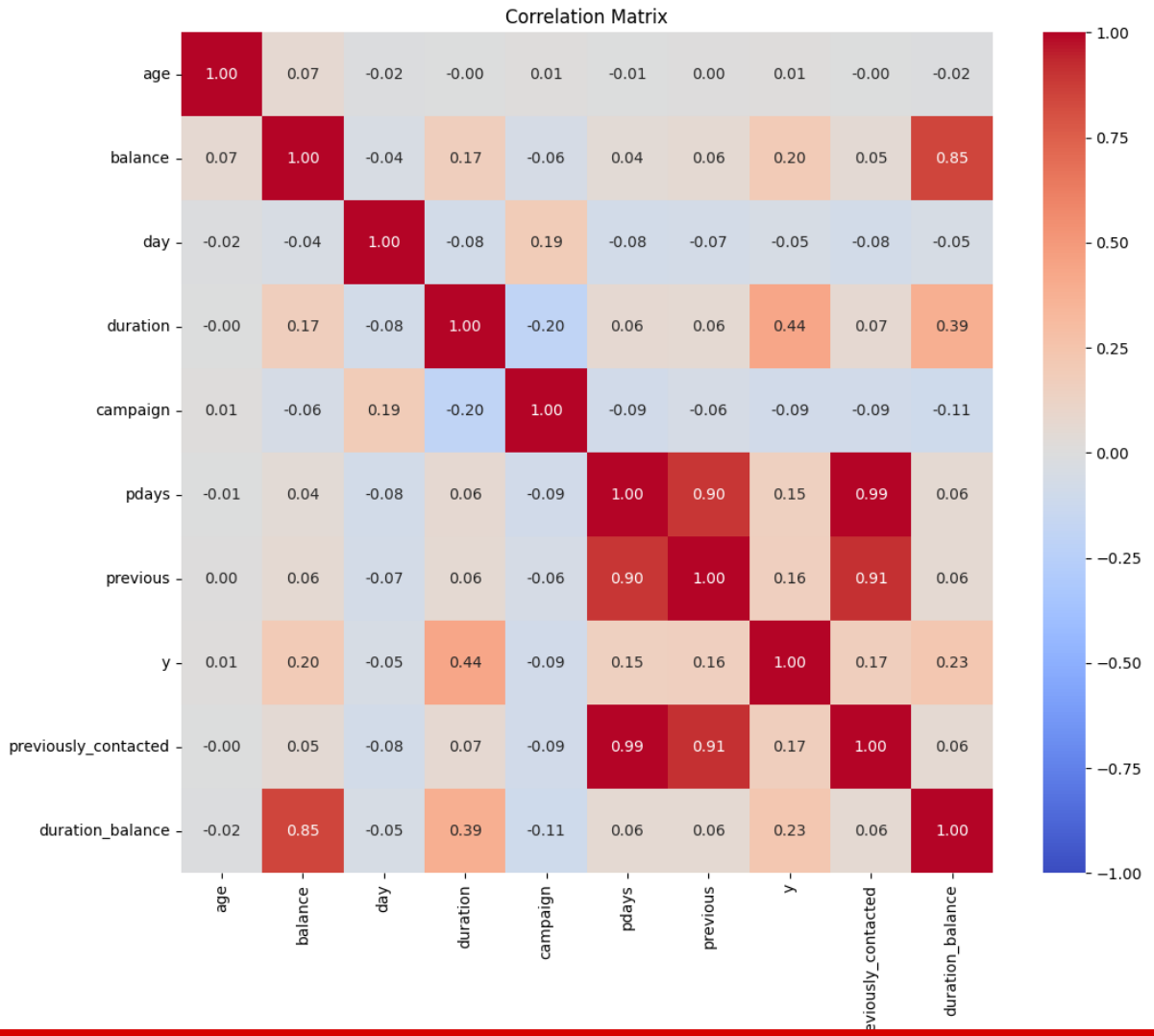
- **Target variable (y):** 0 = No subscription, 1 = Subscription.
- Large variety of categorical and numerical features.

Data Exploration

Steps:

1. Checked dataset shape and structure
2. Verified missing values
3. Studied categorical distributions
4. Studied numerical distributions and skewness

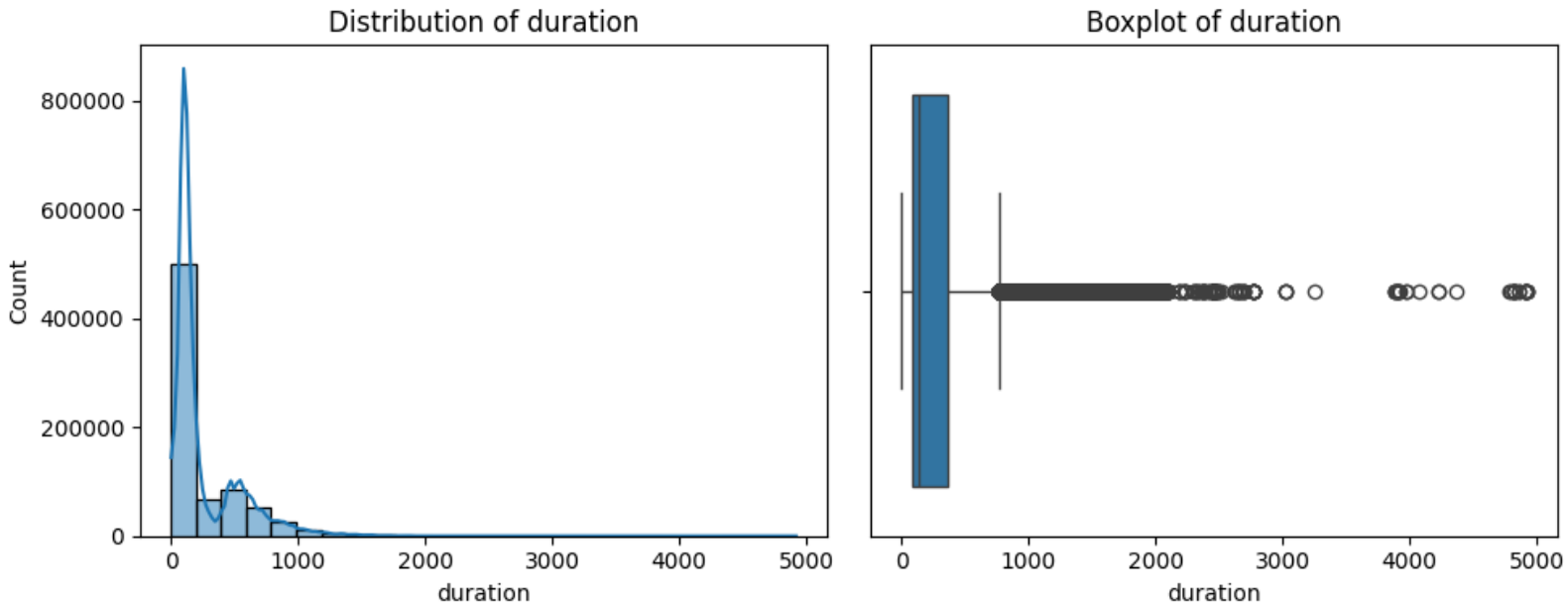
Correlation Analysis



Skewed Variables

Some variables were highly skewed:

- previous contacts
- balance
- campaign
- duration



Feature Engineering

Ideas implemented:

- Transformation for skewed data (either cube root or log1p)
- Binary feature for previous contact (pdays == -1)
- Interaction feature:
  - job x education
  - duration x balance
- Encoding categorical variables

Encoding Strategy

Several approaches were tested (with similar success):

- Label Encoding
- Frequency Encoding
- Target Encoding
- Models' native category handling

Models Tested

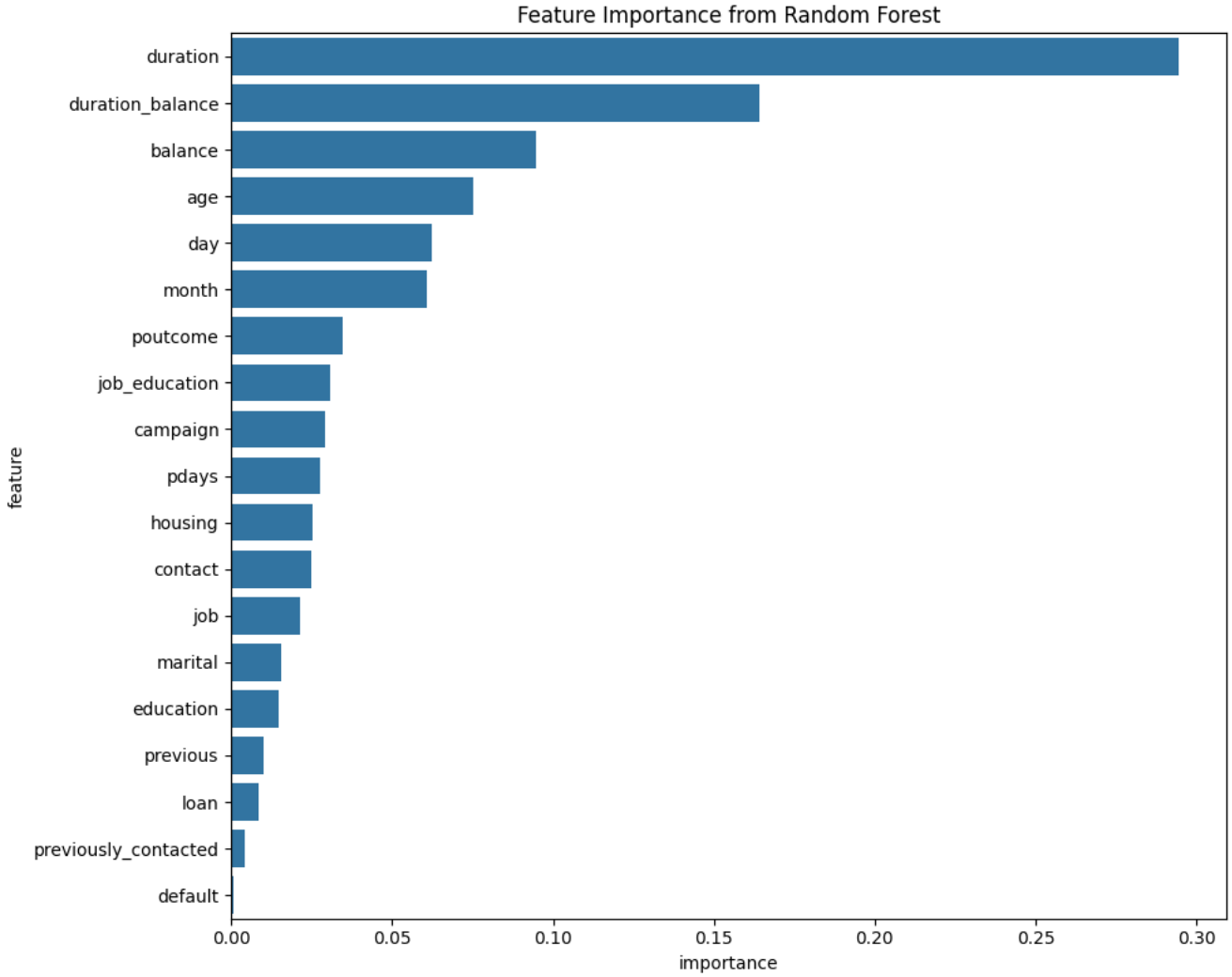
- Random Forest baseline
- **XGBoost** (level-wise horizontal growth: more balanced, less overfitting)
- **LightGBM** (leaf-wise vertical growth: faster loss reduction but potential overfitting)
- **CatBoost** (uses symmetric trees to reduce overfitting)

All evaluated with Stratified 10-fold CV.

CatBoost got a major boost when it received string (categorical) features to handle natively

For all models, a small random hyperparameter search was carried out

Feature Importance



Model Ensemble

Stacking method:

- Combine OOF predictions from:
  - XGBoost
  - CatBoost
  - LightGBM
- **Meta-learner**: Logistic Regression

Improves robustness and performance. It acts as a "blender", weighting the predictions of the base models to optimize final accuracy, often outperforming complex models on their own.

Evaluation Metric

Metric used:

- ROC-AUC score

Cross-validation ensures:

- No overfitting
- Reliable leaderboard performance

Results

Final Score:

- ROC-AUC = 0.97478
- Top 14% of leaderboard

Main improvements came from:

- Feature engineering
- Ensemble learning

Lessons Learned

- Feature engineering is crucial to increase AUC
  - Combining certain features even when the new variables are highly correlated
  - Choosing the right encoding strategy
  - Tuning the model parameters
- Tree models perform well on tabular data
- Ensembling improves results

References

- Kaggle Bank Marketing Competition
- Scikit-Learn documentation
- XGBoost / LightGBM / CatBoost papers

Thank You!